

Energy Derivatives: Pricing and Risk Management

Energy Derivatives: Pricing and Risk Management

Les Clewlow and Chris Strickland



Published in 2000 by Lacima Publications
London, England
Email: contact@lacimagroup.com
Website: www.lacimagroup.com

Copyright © Les Clewlow and Chris Strickland

All rights reserved. No part of this publication may be reproduced, copied, stored in a retrieval system, or transmitted in any form or by any means, electronic, mechanical, photocopying, recording, or otherwise without either prior written permission from the publisher, the copyright owner, or a licence permitting restricted copying in the United Kingdom issued by the Copyright Licensing Agency Ltd, 90 Tottenham Court Road, London, UK W1P 0LP.

ISBN 0-9538896-0-2

Typeset by Alden Bookset, Oxford
Front cover photography by Anthony Marsland
Printed and bound in Great Britain by Biddles Limited

Contents

ABOUT THE AUTHORS	x
PREFACE	xii
ACKNOWLEDGEMENTS	xv
CHAPTER 1 INTRODUCTION TO ENERGY DERIVATIVES AND FUNDAMENTALS OF MODELLING AND PRICING	1
1.1 INTRODUCTION TO ENERGY DERIVATIVES	1
1.2 FUNDAMENTALS OF MODELLING AND PRICING	6
1.3 NUMERICAL TECHNIQUES	9
1.3.1 The Trinomial Method	00
1.3.2 Monte Carlo Simulation	00
1.4 SUMMARY	00
CHAPTER 2 UNDERSTANDING AND ANALYSING SPOT PRICES	00
2.1 INTRODUCTION	00
2.2 MEAN REVERSION	00
2.3 SIMULATING MEAN REVERSION	00
2.4 THE HALF-LIFE OF A MEAN REVERTING PROCESS	00
2.5 STOCHASTIC VOLATILITY	00
2.6 SIMULATING STOCHASTIC VOLATILITY	00
2.7 JUMPS IN THE SPOT PRICE	00
2.8 SIMULATING JUMPS IN THE SPOT PRICE	00
2.9 ESTIMATION OF THE MEAN REVERSION RATE	00
2.10 INTUITIVE ESTIMATION OF JUMP PROCESS PARAMETERS	00
2.11 MAXIMUM LIKELIHOOD ESTIMATION OF JUMP PARAMETERS	00
2.12 INCORPORATING SEASONAL PATTERNS INTO THE MODELS	00
2.13 CONCLUSIONS	00
CHAPTER 3 VOLATILITY ESTIMATION IN ENERGY MARKETS.	00
3.1 INTRODUCTION	00
3.2 ESTIMATING VOLATILITY	00
3.2.1 Estimation of Volatility From Historical Data	00
3.2.2 Estimation of Volatility For a Mean Reverting Process	00
3.2.3 Volatility Estimation: Special Issues	00
3.2.4 Intraday Price Variability	00

Contents

3.2.5	Estimation of Volatility for a Basket	00
3.2.6	Implied Volatility	00
3.3	STOCHASTIC VOLATILITY MODELS	00
3.3.1	Estimation and Testing	00
3.3.2	Ordinary Least Squares	00
3.3.3	Maximum Likelihood	00
3.3.4	Testing	00
3.3.5	Examples From the Energy Commodity Markets	00
3.4	SUMMARY	00
CHAPTER 4	ENERGY FORWARD CURVES	00
4.1	INTRODUCTION	00
4.2	CONSTRUCTING FORWARD CURVES	00
4.2.1	Cost of Carry Relationship	00
4.2.2	Forward Price Bounds for Energies	00
4.2.3	Seasonality in Prices	00
4.3	FORWARD CURVES IN THE ELECTRICITY MARKET	00
4.3.1	Arbitrage Pricing Approach	00
4.3.2	The Econometric Approach	00
4.3.3	The Spot Price Modelling Approach	00
4.4	SUMMARY	00
CHAPTER 5	ENERGY DERIVATIVES STRUCTURES AND APPLICATIONS	00
5.1	INTRODUCTION	00
5.2	EXCHANGE TRADED INSTRUMENTS	00
5.3	SWAPS	00
5.3.1	Vanilla Swap	00
5.3.2	Variable Volume Swap	00
5.3.3	Differential Swap	00
5.3.4	Margin or Crack Swap	00
5.3.5	Participation Swap	00
5.3.6	Double-Up Swap	00
5.3.7	Extendable Swap	00
5.4	CAPS, FLOORS AND COLLARS	00
5.5	SWAPTIONS	00
5.6	COMPOUND OPTIONS – CAPTIONS AND FLOPTIONS	00
5.7	SPREAD AND EXCHANGE OPTIONS	00
5.7.1	Calendar Spreads	00
5.7.2	Crack Spreads	00
5.7.3	Exchange Options	00
5.8	PATH-DEPENDENT OPTIONS	00
5.8.1	Asian Options – Average Price and Average Strike	00
5.8.2	Barrier Options	00
5.8.3	Lookback Options – Fixed Strike and Floating Strike	00
5.8.4	Ladder and Cliquet Options	00
5.9	SUMMARY	00

CHAPTER 6 SPOT PRICE MODELS AND PRICING STANDARD INSTRUMENTS	00
6.1 INTRODUCTION	00
6.2 SINGLE FACTOR MODELS	00
<i>Futures and Forward Pricing</i>	00
<i>Option Pricing</i>	00
<i>The Schwartz Single Factor Model</i>	00
<i>Futures and Forward Pricing</i>	00
<i>Option Pricing</i>	00
6.3 TWO FACTOR MODELS	00
<i>Futures and Forward Pricing</i>	00
<i>Option Pricing</i>	00
<i>The Schwartz 1 Factor Approximation</i>	00
<i>Option Pricing in the Long Term Model</i>	00
6.4 THREE FACTOR MODELS	00
<i>Futures and Forward Pricing</i>	00
<i>Option Pricing</i>	00
6.5 CHOOSING A SPOT PRICE MODEL	00
6.6 SUMMARY	00
CHAPTER 7 SPOT PRICE MODELS: PRICING PATH DEPENDENT AND AMERICAN STYLE OPTIONS	00
7.1 INTRODUCTION	00
7.2 SIMULATION METHODS FOR ENERGY SPOT PRICE MODELS	00
7.2.1 Single Factor Simulation	00
7.2.2 Example – Valuing Spread Options	00
7.2.3 Example – Valuing European Swaptions	00
7.2.4 Incorporating Seasonality into Monte Carlo Simulations	00
7.2.5 Computation of Hedge Sensitivities	00
7.2.6 Simulation of Multi-Factor Models	00
7.2.7 Generation of Random Numbers	00
7.2.8 Valuing Path Dependent Options Using Simulation	00
7.3 BUILDING TRINOMIAL TREES FOR THE ENERGY SPOT PROCESS	00
7.4 IMPLIED TRINOMIAL TREES	00
7.5 PRICING GENERAL PATH DEPENDENT ENERGY OPTIONS IN SPOT PRICE TREES	00
7.5.1 Pricing Asian Energy Options in a Trinomial Tree	00
7.5.2 Pricing Swing Options	00
7.6 SUMMARY	00
CHAPTER 8 SPOT PRICE MODELS: PRICING PATH DEPENDENT AND AMERICAN STYLE OPTIONS	00
8.1 INTRODUCTION	00
8.2 A SIMPLE MODEL FOR THE FORWARD CURVE	00
8.3 THE DYNAMICS OF THE FORWARD CURVE	00
8.4 A GENERAL MULTI-FACTOR MODEL OF THE FORWARD CURVE	00
8.5 RELATIONSHIP BETWEEN THE FORWARD CURVE AND SPOT PRICE MODELS	00

Contents

8.6	ESTIMATION OF THE VOLATILITY FUNCTIONS	00
8.6.1	Historical Estimation of the Forward Curve Volatility Functions	00
8.6.2	Example Analysis of NYMEX Crude Oil Futures	00
8.6.3	Incorporating Seasonality into Volatility Functions	00
8.7	PRICING STANDARD EUROPEAN OPTIONS	00
8.7.1	Options on Spot	00
8.7.2	Options on Forward Contracts	00
8.8	PRICING GENERAL EUROPEAN CONTINGENT CLAIMS	00
	<i>Example: Pricing a European Energy Swaption</i>	00
8.9	PRICING EXOTIC OPTIONS	00
8.9.1	Calendar Spread Options	00
8.9.2	Crack Spread Options	00
8.9.3	Asian Options	00
8.9.4	Americian Options in Forward Curve Models	00
8.10	IMPLIED ESTIMATION OF THE VOLATILITY FUNCTIONS	00
8.10.1	Handling Volatility Skews and Smiles	00
8.11	SUMMARY	00
CHAPTER 9	RISK MANAGEMENT OF ENERGY DERIVATIVES	00
9.1	INTRODUCTION	00
9.2	DELTA HEDGING OF ENERGY DERIVATIVE POSITIONS	00
9.3	GAMMA HEDGING	00
9.4	HEDGING VOLATILITY AND OTHER SOURCES OF MARKET RISK	00
9.5	FACTOR HEDGING	00
9.6	SUMMARY	00
CHAPTER 10	RISK MANAGEMENT OF ENERGY DERIVATIVES	00
10.1	INTRODUCTION	00
10.2	WHAT IS VALUE AT RISK?	00
10.3	MODELLING THE MARKET VARIABLES	00
10.4	FORECASTING STANDARD DEVIATIONS (VOLATILITY)	00
10.5	FORECASTING COVARIANCES AND CORRELATIONS	00
10.6	VALUE AT RISK METHODOLOGIES	00
10.6.1	Variance-Covariance (Delta VaR)	00
10.6.2	Delta-Gamma VaR	00
10.6.3	Monte Carlo Simulation	00
10.6.4	Historical Simulation	00
10.7	TESTING VALUE-AT-RISK ESTIMATES	00
10.7.1	Testing the VaR Forecast for Market Variables	00
10.8	CONCLUSIONS	00
CHAPTER 11	CREDIT RISK IN ENERGY MARKETS	00
11.1	INTRODUCTION	00
11.2	WHAT IS CREDIT RISK?	00
11.3	THE KEY COMPONENTS OF CREDIT RISK MODELS	00
11.3.1	Credit Ratings and Credit Rating Transition Probabilities	00
11.3.2	Credit Spreads	00

Contents

11.3.3	Recovery Rates in Default	00
11.3.4	A Summary of Major Credit Risk Models	00
11.4	CREDITMETRICS	00
11.4.1	Modelling Credit Rating Changes	00
11.4.2	Simulation of Counterparty Market Returns	00
11.4.3	Credit Spreads, Recovery Rates and the Calculation of Derivative Values	00
11.4.4	Modelling Recovery Rates	00
11.4.5	Calculation of the Credit Risk for an Energy Derivative Portfolio	00
11.5	CREDITRISK +	00
11.5.1	Modelling the Occurrence of Defaults	00
11.6	CREDIT RISK MEASURES AND MANAGEMENT	00
11.6.1	Marginal Risk	00
11.6.2	Concentration Risk	00
11.6.3	Managing Credit Risk	00
11.6.4	Assessing the Models	00
11.7	CONCLUSIONS	00
REFERENCES		00
INDEX		00

About the Authors

Les Clewlow is a Principal of Lacima Group which has been supplying training, software and consulting to financial, commodity and energy institutions and software companies worldwide for more than ten years. He is an Associate Research Fellow at the School of Finance and Economics, University of Technology Sydney and the Financial Options Research Centre, University of Warwick in the United Kingdom. He has a First Class Honours degree in Physics with Astrophysics (Birmingham), and a PhD in Computer Science (Warwick). He has recent publications in Risk Magazine, Energy and Power Risk Management and Journal of Economic Dynamics and Control. His current interests include energy and commodity risk management, pricing and hedging exotic options, interest rate models and numerical algorithms.

Chris Strickland co-founded and is principal of Lacima Group. He is an Associate Research Fellow at the School of Finance and Economics, University of Technology Sydney and the Financial Options Research Centre, University of Warwick in the United Kingdom, and an associate researcher at the Instituto de Estudios Superiores de Administración, Caracas, Venezuela. Previously, Chris worked for RBC Gilts Ltd, and Kitcat and Aitken. & Co. in London. He is the co-author, with Les Clewlow, of the book *“Implementing Derivatives Models”* and co-editor of the book *“Exotic Options: The State of the Art”*. He has a First Class Honours degree in Pure Mathematics (Liverpool), an MSc in Mathematics (Warwick) and a PhD in Finance (Warwick). His current interests include energy derivative pricing, interest rate models and numerical algorithms.

Vincent (Vince) Kaminski is Managing Director and Head of Research for Enron Corp. Vince joined Enron in June of 1992. Previously, Vince was a vice president in the research department of Salomon Brothers in New York (Bond Portfolio Analysis Group) and a manager in AT&T Communications (Long Lines) in Bedminster, New Jersey. In his current position, Vince is responsible for development of analytical tools for pricing of commodity options and other commodity transactions, hedging strategies, optimisation of financial and physical transactions, as well as development of value-at-risk systems. Enron Corp. Research manages the largest portfolio of fixed-price and derivative natural gas contracts in the world and has invested significant time and effort in the development of state-of-the-art risk management systems. Vince is a recipient of the 1999 James H. McGraw Award for Energy Risk Management (Energy Risk

About the Authors

Manager of the Year). Vince holds an M.S. degree in international economics, a Ph.D. degree in mathematical economics from the Main School of Planning and Statistics in Warsaw, Poland, and an MBA from Fordham University in New York.

Grant Masson is a vice president with the research group of Enron Corp. He oversees the quantitative support for asset and derivative structure valuations and market analysis for both domestic and international electricity trading and origination. He is also responsible for quantitative support for corporate Value- and Credit-at-Risk measurement systems. Prior to joining Enron in 1994, Grant spent five years at the University of Basel as a research scientist specialising in experimental nuclear physics. He received a BA from Rice University in Houston and MS and Ph.D. degrees in physics from the University of Wisconsin – Madison.

Ronnie Chahal is a manager in the Enron Corp. Research group. He has a Ph.D. in Finance from Georgia State University, Atlanta, Georgia. He currently provides analytical support to the risk management group in Enron Energy Services.

Preface

One of our main objectives in writing *Energy Derivatives: Pricing and Risk Management* has been to bring together as many of the various approaches for the pricing and risk management of energy derivatives as possible, to discuss in-depth the models, and to show how they relate to each other. In this way, we hope to help the reader to analyse the different models, price a wide range of energy derivatives, or to build a risk management system which uses a consistent modelling framework. We believe that for practitioners this last point is very important and we continue to stress in our articles and presentations the dangers of using flawed risk management and pricing systems. Using ad-hoc and inconsistent models for different instruments and markets gives arbitrage opportunities to your competitors. However, it is not our wish to concentrate on one particular model or models, to the exclusion of the others because we believe that the choice should rest with the user (although it will probably be clear from our discussions which model(s) we prefer). We therefore try and give as clear account as possible of the advantage and disadvantages of all the models so that the reader can make an informed choice as to the models which best suit their needs.

In order to meet our objectives, the book is divided into 11 chapters. In chapter 1 we give an overview of the fundamental principles needed to model and price energy derivatives which will underpin the remainder of the book. In addition to introducing the techniques that underlie the Black-Scholes modelling framework, we outline the numerical techniques of trinomial trees and Monte Carlo simulation for derivative pricing, which are used throughout the book.

In Chapter 2 we discuss the analysis of spot energy prices. As well as analysing empirical price movements, we propose a number of processes that can be used to model the prices. We look at the well-know process of Geometric Brownian Motion as well as mean reversion, stochastic volatility and jump processes, discussing each and showing how they can be simulated and their parameters estimated.

Chapter 3, written by Vince Kaminski, Grant Masson and Ronnie Chahal of Enron Corp., discusses volatility estimation in energy commodity markets. This chapter builds on the previous one. It examines in detail the methods, merits and pitfalls of the volatility estimation process assuming different pricing models introduced in chapter 2. Examples from crude, gas, and electricity markets are used to illustrate the technical and interpretative aspects of calculating volatility.

Chapter 4 examines forward curves in the energy markets. Although such curves are well understood and straight-forward in many financial and commodity markets, the

difficulty of storage in many energy markets leads to less well defined curves. In this chapter we describe forward price bounds for energy prices and the building of forward curves from market instruments. We outline the three main approaches which have been applied to building forward curves in energy markets; the arbitrage approach, the econometric approach, and deriving analytical values by modelling underlying stochastic factors.

Chapter 5 presents an overview of structures found in the energy derivative markets and discusses their uses. Examples of products analysed in this chapter include a variety of swaps, caps, floors and collars, as well as energy swaptions, compound options, Asian options, barrier options, lookback options, and ladder options.

Chapter 6 investigates single and multi-factor models of the energy spot price and the pricing of some standard energy derivatives. Closed form solutions for forward prices, forward volatilities, and European option prices (both on the spot and forwards) are derived and presented for all the models in this chapter including a three factor, stochastic convenience yield and interest rate model.

Chapter 7 shows how the prices of path dependent and American style options can be evaluated for the models in Chapter 6. Simulation schemes are developed for the evaluation of European style options and applied to a variety of path dependent options. In order to price options which incorporate early exercise opportunities, a trinomial tree approach is developed. This tree is built to be consistent with the observed forward curve and can be used to price exotic as well as standard European and American style options.

Chapter 8 describes a methodology for valuing energy options based on modelling the whole of the market observed forward curve. The approach results in a multi-factor model that is able to capture realistically the evolution of a wide range of energy forward curves. The user defined volatility structures can be of an extremely general form. Closed-form solutions are developed for pricing standard European options, and efficient Monte Carlo schemes are presented for pricing exotic options. The chapter closes with a discussion of the valuation of American style options.

Chapter 9 focuses on the risk management of energy derivative positions. In this chapter we discuss the management of price risk for institutions that trade options or other derivatives and who are then faced with the problem of managing the risk through time. We begin with delta hedging a portfolio containing derivatives and look at extensions to gamma hedging – illustrating the techniques using both spot and forward curve models. The general model presented in Chapter 8 is ideally suited to multi-factor hedging of a portfolio of energy derivatives which is also discussed.

Chapter 10 examines the key risk management concept of Value at Risk (VaR) applied to portfolios containing energy derivative products. After discussing the concept of the measure, we look at how the key inputs (volatilities, covariances, correlations, etc) can be estimated. We then compare the four major methodologies for computing VaR, delta, delta-gamma, historical simulation and Monte-Carlo simulation, applying each to the same portfolio of energy options. In this chapter we also look at testing the VaR estimates for various underlying energy market variables.

Preface

Finally, in Chapter 11 we review modelling approaches to credit risk. We look in detail at two quite different approaches, *CreditMetrics* (J. P. Morgan (1997)) and *CreditRisk+* (Credit Suisse Financial Products (1997)), for which detailed information is publicly available. Together, these provide an extensive set of tools with which to measure credit risk. We present numerical examples of applying these techniques to energy derivatives.

We must stress that the models and methods we present in this book are tools, which should be used with the benefit of an understanding of how both the ‘tool’ and the market works. The techniques we describe are certainly not “magic wands” which can be waved at data and risk management problems to provide instant and perfect solutions. To quote from the RiskMetrics Technical Document “. . . no amount of sophisticated analytics will replace experience and professional judgement in managing risk”. However, the right tools correctly used make the job a lot easier!

We are, as always, pleased to receive any comments on the book – please use the dedicated email address: energyderivatives@lacimagroup.com.

Les Clewlow, Lacima Group
Chris Strickland, Lacima Group
Vince Kaminski, Enron Corp. Research
August 2000

Acknowledgements

It is a pleasure for us to acknowledge numerous colleagues, both academic and practitioner, for the many useful discussions that have contributed to our knowledge over the years, as well as numerous friends, all of whom have helped us in the production of this book. To those we fail to mention by name, please accept our apologies;

Peter Ash, Carl Chiarella, Brian Collins, Nigel Collins, Gary Cox, Olivier Croissant, Nadima El-Hassan, Doug Gardner, Tony Hall, Simon Hurst, Chris Leeds, Vincent Kaminski, Adam Kucera, Andrew Jameson, Sandra Jackson, Rob de Rozario, Jamie Rogers, Peter Israel, Glenn Kentwell, John Martin, Grant Masson, Ian Merker, Sudip Mukherjee, Bernard Murphy, Christina Nikitopoulos, Mike Roffey, Mark Schneider, Vito Scoppio, Gary Stanford, Max Stevenson, Steve Thomas, Louise Walker, Amanda Walker and Joe Winsen.

Although too numerous to mention individually, participants at Lacima's energy related courses in Sydney, Melbourne, Houston, New York, London, and Brazil have helped to clarify our ideas and provided us with data and examples.

Special thanks go to Michel Booth, Ilia Bouchiev, Alexander Edyeland, Mark Garman, Corwin Joy, and David Shimko who read the original manuscript and gave us extensive feedback which led to significant improvements in the finished version. Julie Brennan and Rubin Rajendram greatly assisted in the production process.

The authors would also like to acknowledge the financial support of the School of Finance and Economics, University of Technology Sydney, Boral Energy and Société Generale.

LC would like to thank and dedicate this book to his Mum, Dad, brother, sister and grandparents.

CS would like to thank his Mum and Dad and to apologise to Lorna for all the disruptions this text has caused. This is for 'Mugs'.

LC
CS